Hutch-ALTTPR Tracker

User Guide

Version 1.1.0

A Link to the Past Randomizer — Auto-Tracking Tool

Disclaimer: This tracker is provided free of charge and is updated on a best-effort basis when issues are discovered. All source code is available at the GitHub repository linked below.

Source code https://github.com/hutchch/ALTTPR-Tracker
Releases https://github.com/hutchch/ALTTPR-Tracker/releases

Quick Start Guide

For best results, launch software in the following order:

- Launch QUSB2SNES or SNI
- Open the Emulator (RetroArch)
- Launch the game to the title screen
- Launch the tracker components (Items, Map, Timer)
  - All windows should show "Connected" in the bottom bar
- Start the game

Prerequisites

- An emulator compatible with WebSocket memory access:
  - RetroArch v1.22.2 or higher is recommended
- A WebSocket bridge — install ONE of the following:
  - QUSB2SNES (Windows and macOS): https://skarsnik.github.io/QUsb2snes/

    - SNI (Windows and macOS): https://github.com/alttpo/sni/releases
- A Link to the Past v1.0 (JP) .sfc ROM file
- A randomized game seed — generate one at https://alttpr.com/

Installation

Windows

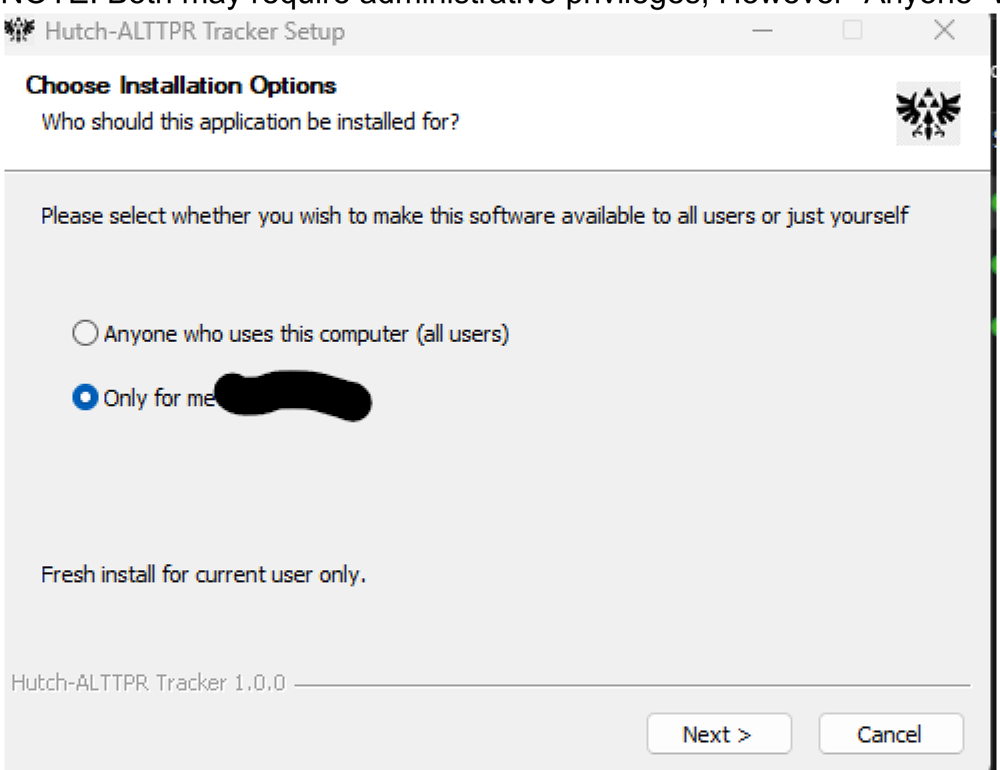
Download the latest release from:

https://github.com/hutchch/ALTTPR-Tracker/releases

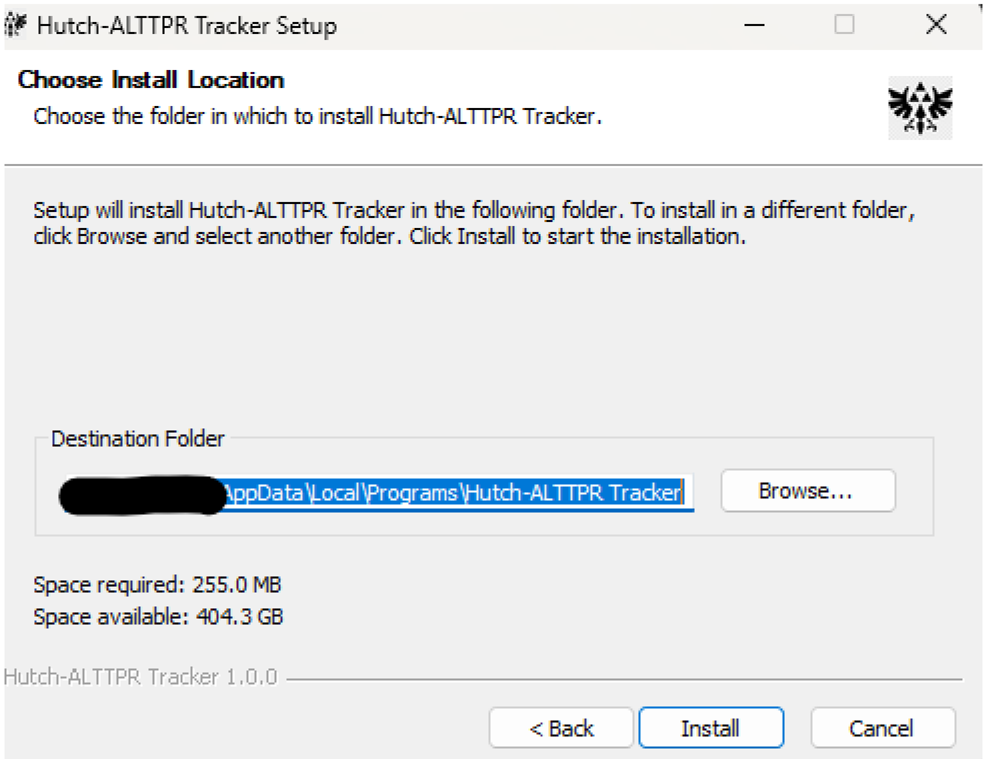
Extract the .zip and run the setup file.

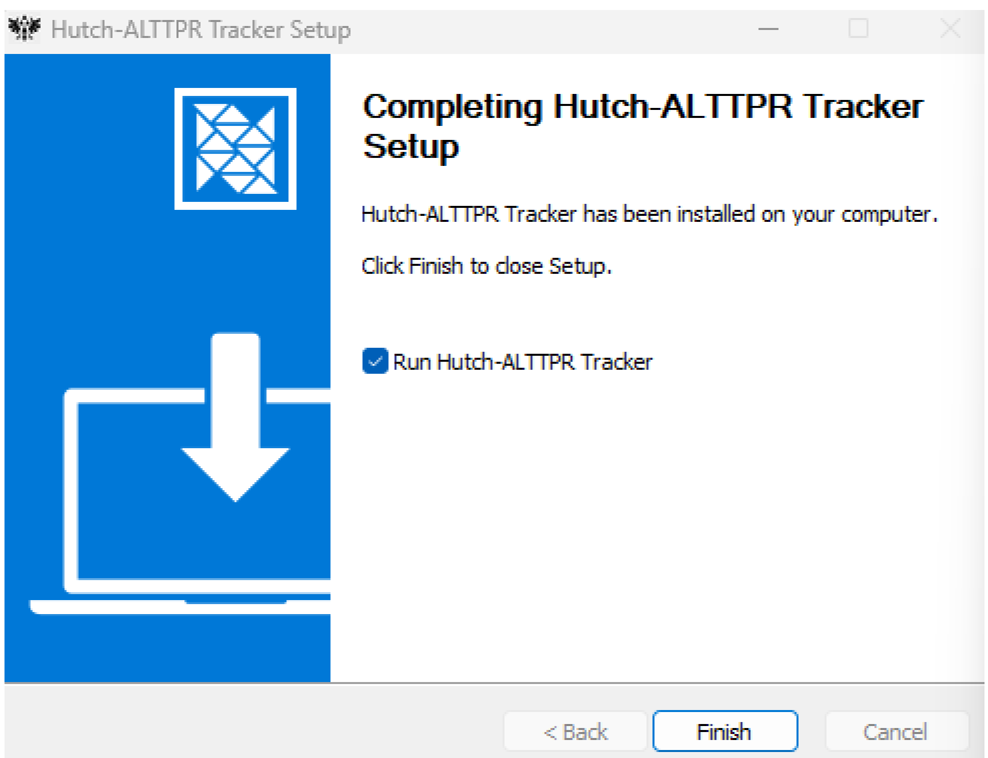
Select either installation option

NOTE: Both may require administrative privileges; However “Anyone” will require them.

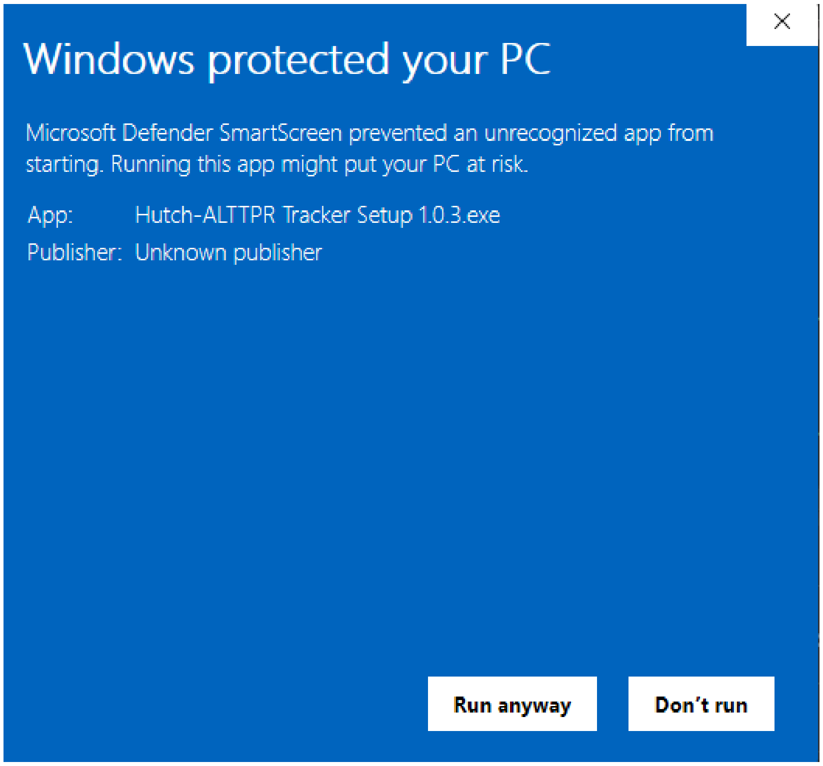


Choose a destination folder and complete the setup.





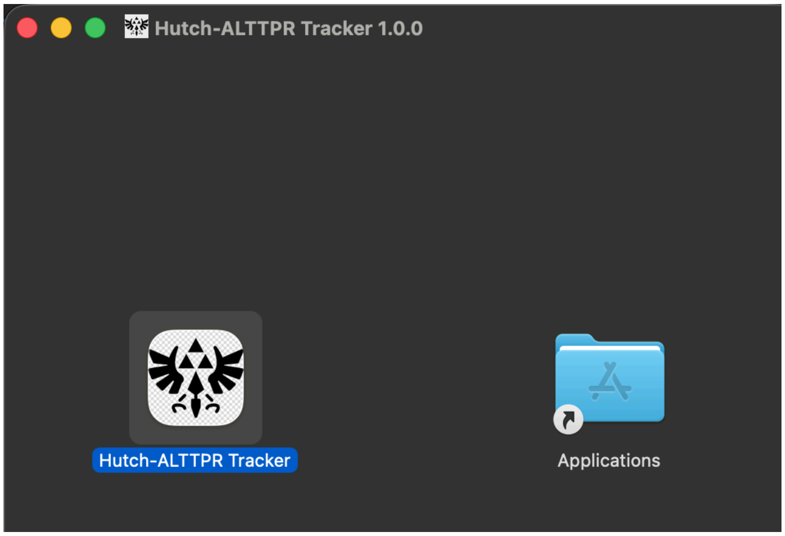
Windows SmartScreen: During install or upgrade, Windows Defender SmartScreen may flag the installer as "unrecognized." Click "More info" then "Run anyway" to proceed. The publisher certificate is an expensive purchase; so all source code is open and available on GitHub.



MacOS

Download the latest release from the same GitHub releases page.
https://github.com/hutchch/ALTTPR-Tracker/releases

Run the .dmg file and drag the application into your Applications folder.

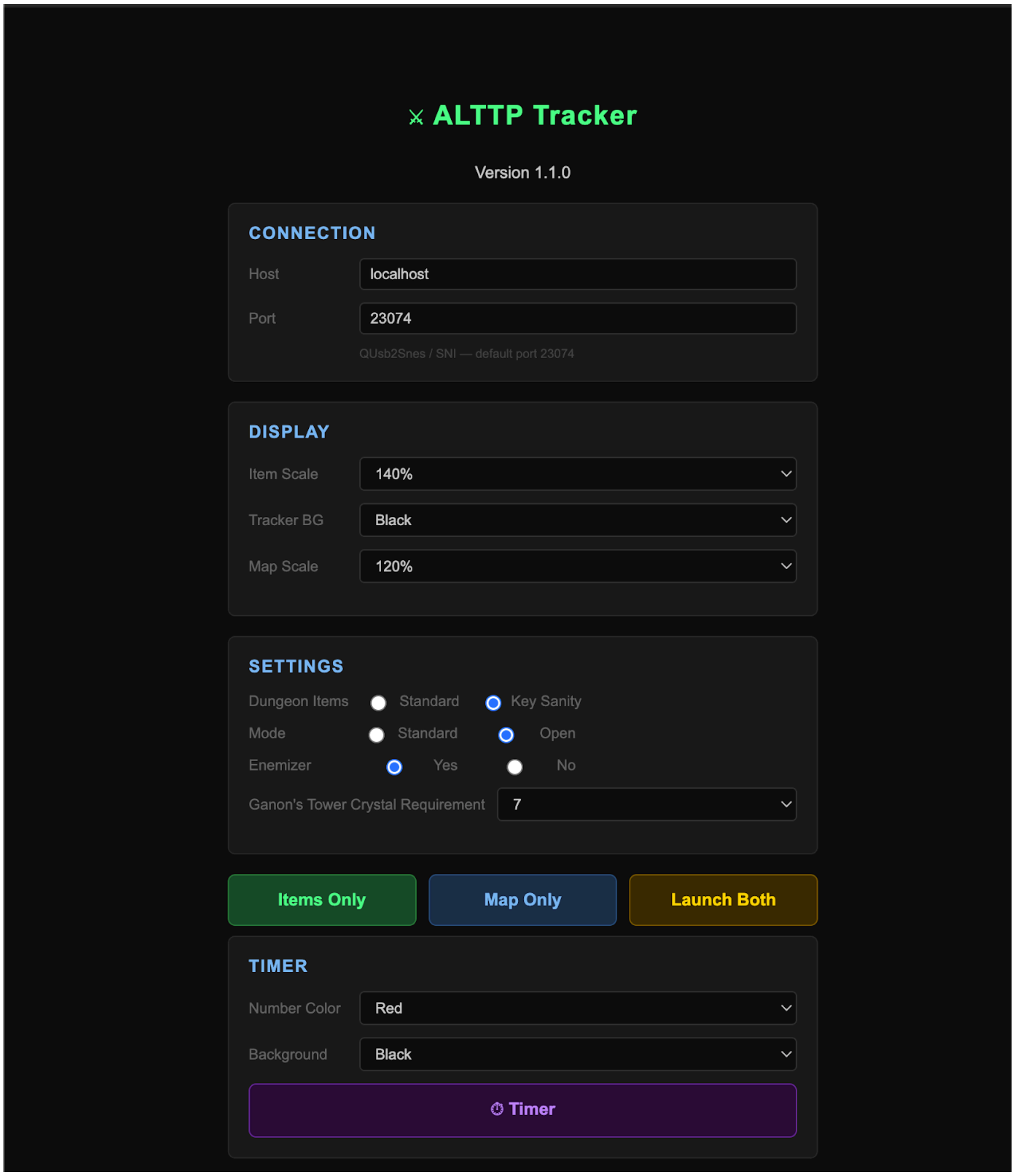


Upgrading

To upgrade to a new version, follow the same steps as a fresh install — the installer will replace the existing version. Your saved settings (host, port, scale, background) are preserved in localStorage and will not be reset.

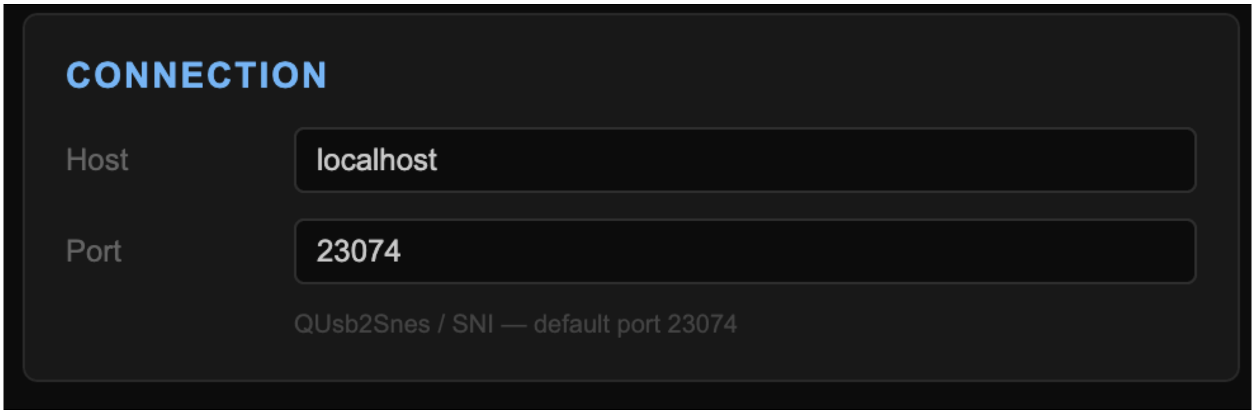
Tracker Launch Settings

When launching, configure the settings to match your generated seed, then press the appropriate launch button. The Items tracker and Map can be launched separately if you need to adjust scaling independently.

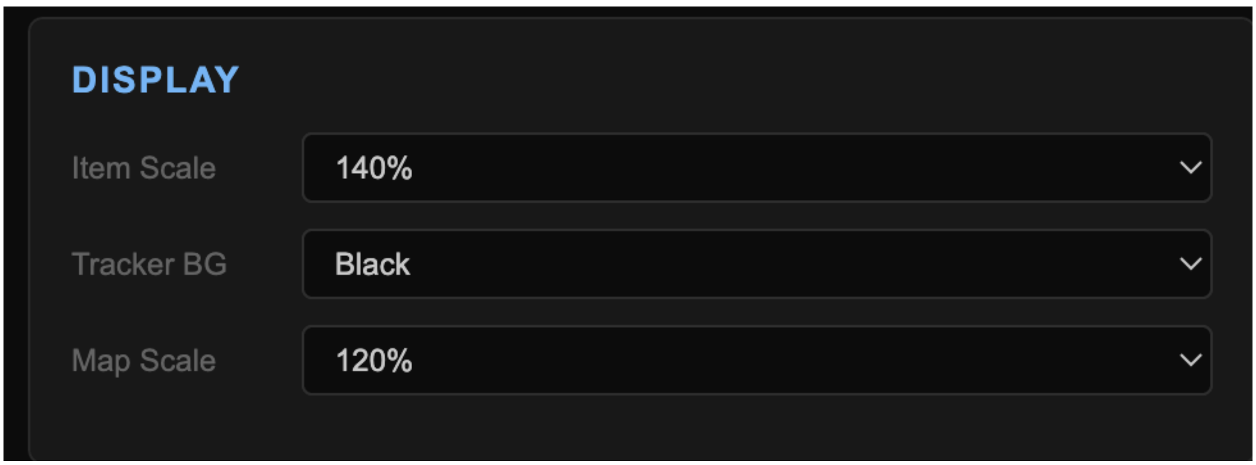


Connection

Enter the host and port for your WebSocket bridge. The default port 23074 works for both SNI and QUSB2SNES. Leave as "localhost" if running everything on the same machine.



Display



Item Scale

Sets the initial zoom level of the Item Tracker window. The scale can also be adjusted at any time using the − and + buttons in the bottom bar of the tracker.

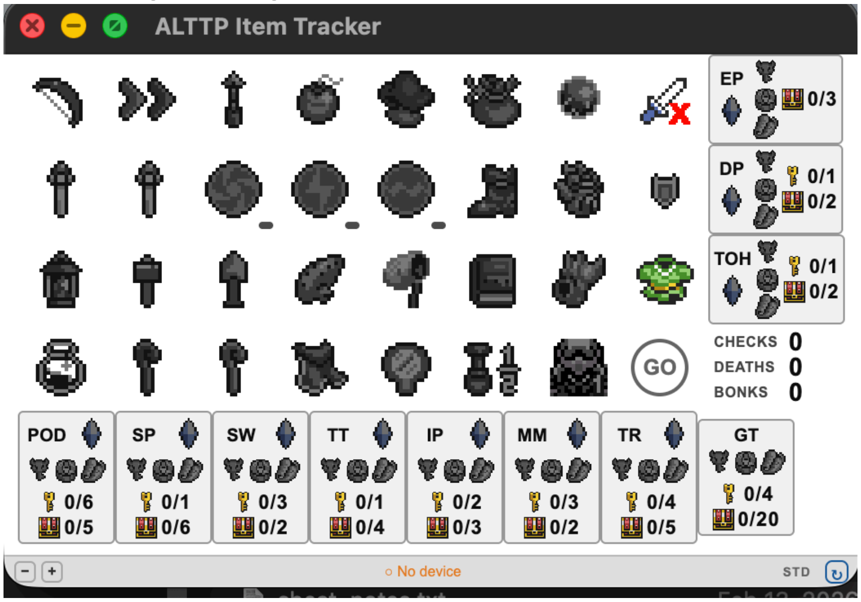
Tracker Background

Sets the background color of the Item Tracker. Four options are available:

- Black — dark background, recommended for most streaming setups

- Grey — medium dark background

- White — light background

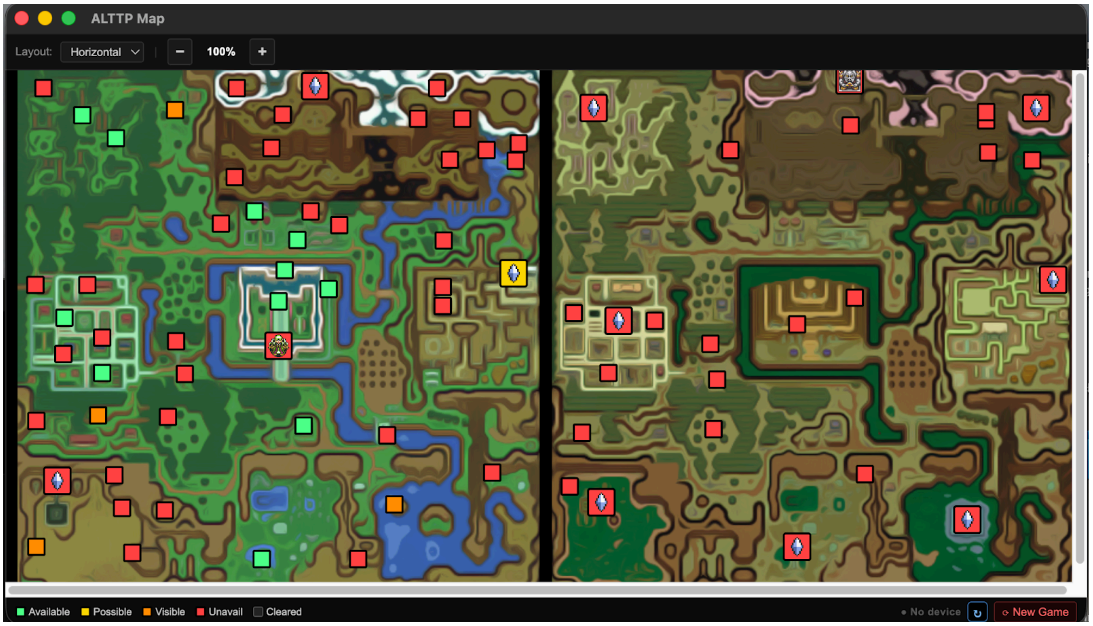


- Transparent — no background; the window chrome is hidden and the tracker floats over your desktop.

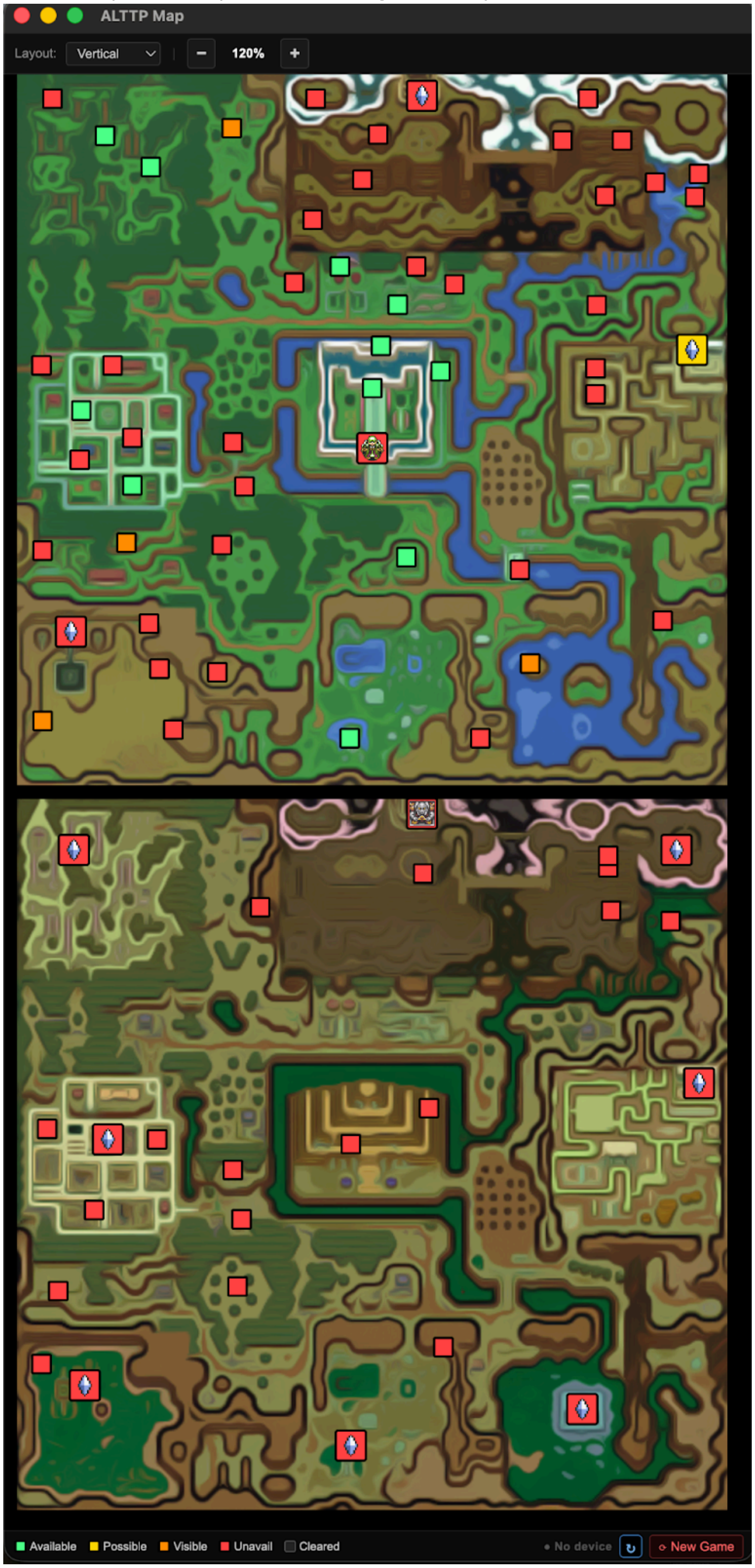
Map Scale

Sets the initial zoom level of the Map window. The map can be rescaled after launch using the − and + buttons in the map's top bar. Layout can also be toggled between

Horizontal (side-by-side)



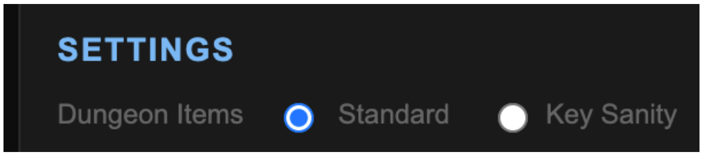
Vertical (stacked) views using the Layout dropdown at the top of the map.



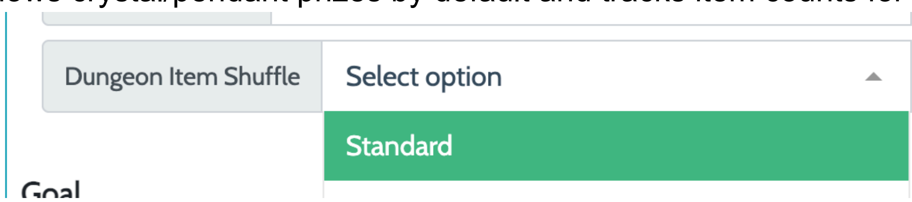
Settings

Dungeon Items

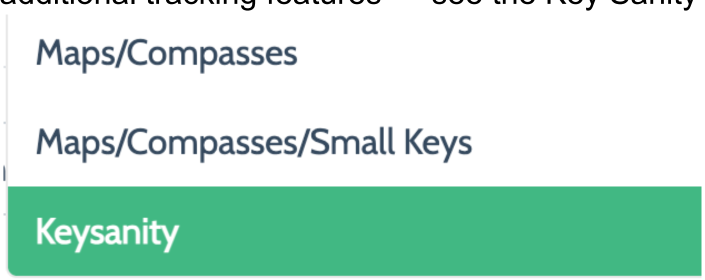
Matches the "Dungeon Item Shuffle" setting from your generated seed. This affects dungeon item counting, prize display, and map logic.



Standard: Use for seeds with Standard dungeon item shuffle from alttpr.com. All dungeon items (maps, compasses, keys) stay in their home dungeons. The tracker shows crystal/pendant prizes by default and tracks item counts for each dungeon.

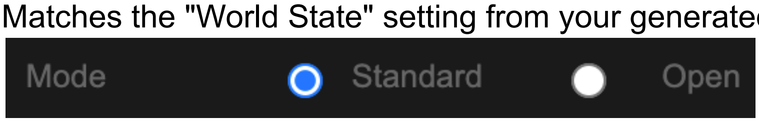


Key Sanity: Use for seeds with Maps/Compasses, Maps/Compasses/Small Keys, or full Keysanity shuffle. Dungeon items are shuffled into the general item pool. Enables additional tracking features — see the Key Sanity section for details.

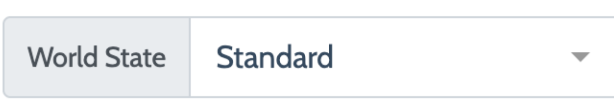


Mode

Matches the "World State" setting from your generated seed.



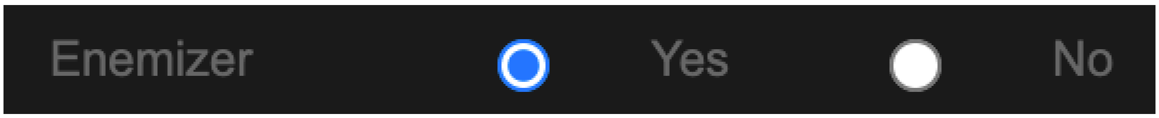
- Standard — the game begins with the Hyrule Castle escape sequence



- Open — the escape is skipped; Link starts directly in the Light World

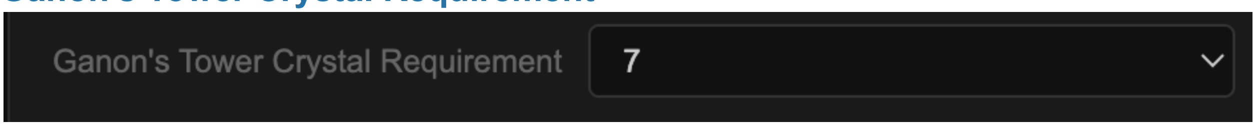
The other “World State” options, when generating the seed, Open mode should be sufficient. See https://alttpr.com/en/options for additional information.

Enemizer



Set to Yes if your seed was generated with Random or Shuffled enemy shuffle. Set to No for seeds with no enemy shuffle (None). This affects Eastern Palace logic — with Enemizer enabled, the bow is not required to access EP.

Ganon's Tower Crystal Requirement



Set this to match the "Open Tower" crystal count from your seed's Goal settings. The map will show GT as green (available) when you have collected at least this many crystals.

Once launched, the tracker should connect. If not, hit the reconnect button in the bottom right corner.

Using the Tracker

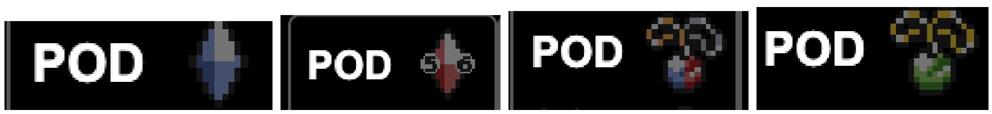
Standard Randomizer

Item Tracking

Items are tracked automatically via SRAM once connected. Click any item to cycle through its states manually — useful if autotracking is unavailable or if you want to mark something ahead of time.

Dungeon Prize Marking

In Standard mode, all crystal dungeons (POD, SP, SW, TT, IP, MM, TR) default to Crystal since it is the most common prize. Click the prize icon on any dungeon slot to cycle through:

- Crystal → Red Crystal → Pendant → Green Pendant → Crystal
- 

The map updates automatically to reflect the correct prize icon. Only change the prizes that differ from the default crystal.

Dungeon Item Counts

Each dungeon slot shows two counters:

- Key icon () — small keys found / total small keys in dungeon
- Chest icon () — non-dungeon items collected / total non-dungeon items

The chest counter excludes maps, compasses, big keys, and small keys from the count — it only tracks items that are meaningful progression checks. In Standard mode the tracker automatically subtracts dungeon items found to keep the count accurate.

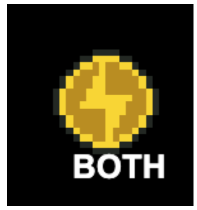
Dungeon Completion Borders

Dungeon slots change border color to indicate completion status:

Border Color	Meaning
Grey (default)	Dungeon not yet complete
Blue (#66f)	Standard mode: all non-dungeon items collected
Green (#2e8b57)	All items collected AND boss prize obtained

Medallion Marking

Right-click the Bombos, Ether, or Quake medallion to mark it as the required medallion for Misery Mire or Turtle Rock. Cycle through: unmarked → MM → TR → BOTH. The map uses this to determine if MM and TR are accessible.

or


Statistics

The stats area (bottom-right of the item tracker) shows three counters that update automatically from SRAM:

- CHECKS (green) — total locations cleared
- DEATHS (red) — total deaths in the current game
- BONKS (orange) — total bonks taken

Go Mode

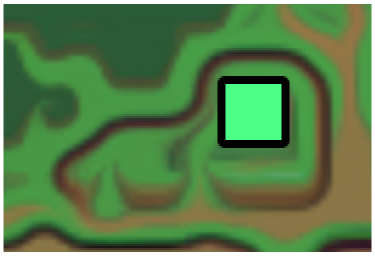
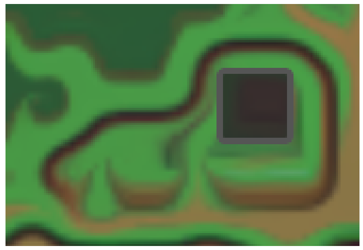
The GO button in the fourth row cycles through states. When activated, the circle fills with the selected color and the text turns black — indicating you have all items needed to complete the game. Inactive state shows a hollow grey circle. Click to toggle through color options.

Map Tracker

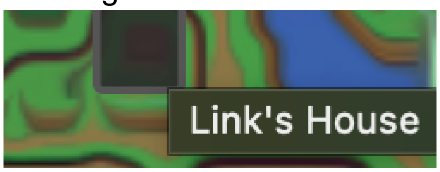
The map window shows all check locations on both the Light World and Dark World maps. Dot colors indicate check availability based on your current items:

Color	Meaning
Green	Available — you have all required items to access the area
Yellow	Possible — accessible but may require dark rooms
Orange	Visible — you can see the item but cannot reach it
Red	Unavailable — missing required items
Dark/Cleared	Already collected

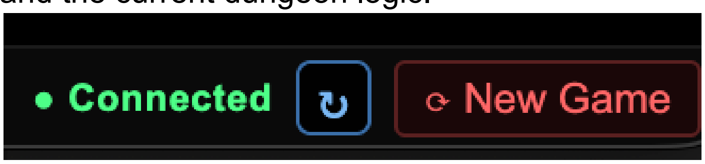
Clicking a dot toggles it as cleared/uncleared manually. The map auto-clears checks when detected via SRAM — so if you manually cleared a check and want to reset the map, click it again to restore the dot.

Hovering over a check dot shows its name.



The "New Game" button in the map's bottom bar resets all check states. Dungeon boxes on the map also update color (green/yellow/red) based on your available items and the current dungeon logic.



Key Sanity Mode

Key Sanity mode activates when the Dungeon Items setting is set to Key Sanity. In this mode, maps, compasses, small keys, and big keys are all shuffled into the general item pool and can be found anywhere in the game.

Item Tracker Changes

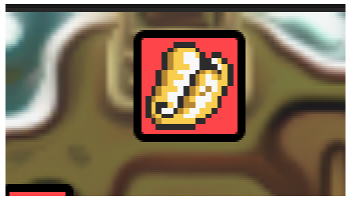
Several changes are visible on the Item Tracker in Key Sanity mode:

Unknown prize (?) by default: All dungeon prizes start as "?" since the prize is unknown until the boss is defeated. Click the prize icon to cycle through: ? → Crystal → Red Crystal → Pendant → Green Pendant → ?.

All item counts visible: In Key Sanity, every chest in a dungeon is a potential item check. The chest counter shows all items found vs total (e.g., 3/6 for Eastern Palace).

Small key count turns green: When all small keys for a dungeon have been found, the key counter (e.g., 6/6) turns green to indicate you have all available keys for that dungeon. This is only shown in Key Sanity mode.

Dungeon map placeholder: When a dungeon's map item is found (tracked via SRAM), the map icon on the dungeon slot lights up. On the map window, if the dungeon map is obtained but the prize has not yet been set from the item tracker, the dungeon box on the map displays the map icon as a placeholder until the prize is manually updated.



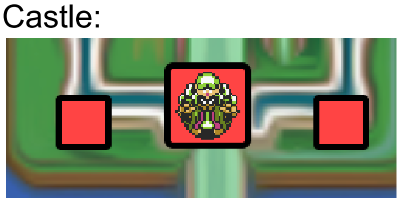
CT Small Key Counter

In Key Sanity mode, a CT key counter appears in the stats area above the CHECKS counter, showing Castle Tower small keys found (0/2). The counter is tracked via SRAM and uses a high-water mark — the count never decreases when keys are used. The counter turns green when both CT keys have been found.

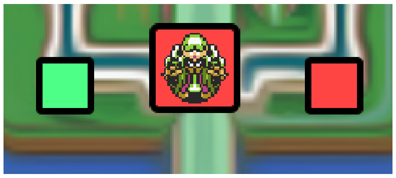
CT Key Addresses: Castle Tower small keys are tracked at SRAM offset 0x4E4, separate from Hyrule Castle/Sewers keys (0x4E1). Only Castle Tower keys count toward the CT 0/2 display.

Castle Tower (CT) Map Logic

In Key Sanity mode, two additional check locations appear on the map near Hyrule Castle:



CT Room 03 (first check): Available with Master Sword (level 2+) or Magic Cape.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CT Dark Maze (second check): Requires lamp + (Master Sword or Cape) + 1 CT small key. Without the small key, the dot shows red even if other requirements are met.

Aga1 dungeon box: The CT dungeon box on the map turns green when you have: lamp + (Master Sword or Cape) + 2 CT small keys. Missing keys shows the box as red.

Dungeon Completion Borders (Key Sanity)

Border behavior changes slightly in Key Sanity mode:

Border Color	Meaning (Key Sanity)
Grey (default)	Dungeon not yet complete
Green (#2e8b57)	All items collected AND boss prize obtained — dungeon fully cleared

NOTE: The blue "all items" border does not apply in Key Sanity mode — since defeating the boss always provides the prize immediately, the green fully-cleared border is the relevant completion indicator.

Once all the keys for a dungeon are collected, the Small Key count will show green

Mode Label

The bottom bar of the Item Tracker shows a small label to the left of the reconnect button indicating the current mode: KS for Key Sanity, STD for Standard. This is a quick reminder of which mode the tracker was launched in.

Timer

The Timer is an optional companion window for tracking run time. It connects to the same WebSocket as the other tracker components

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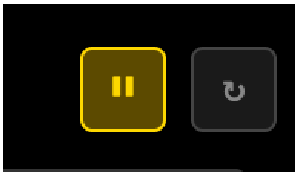
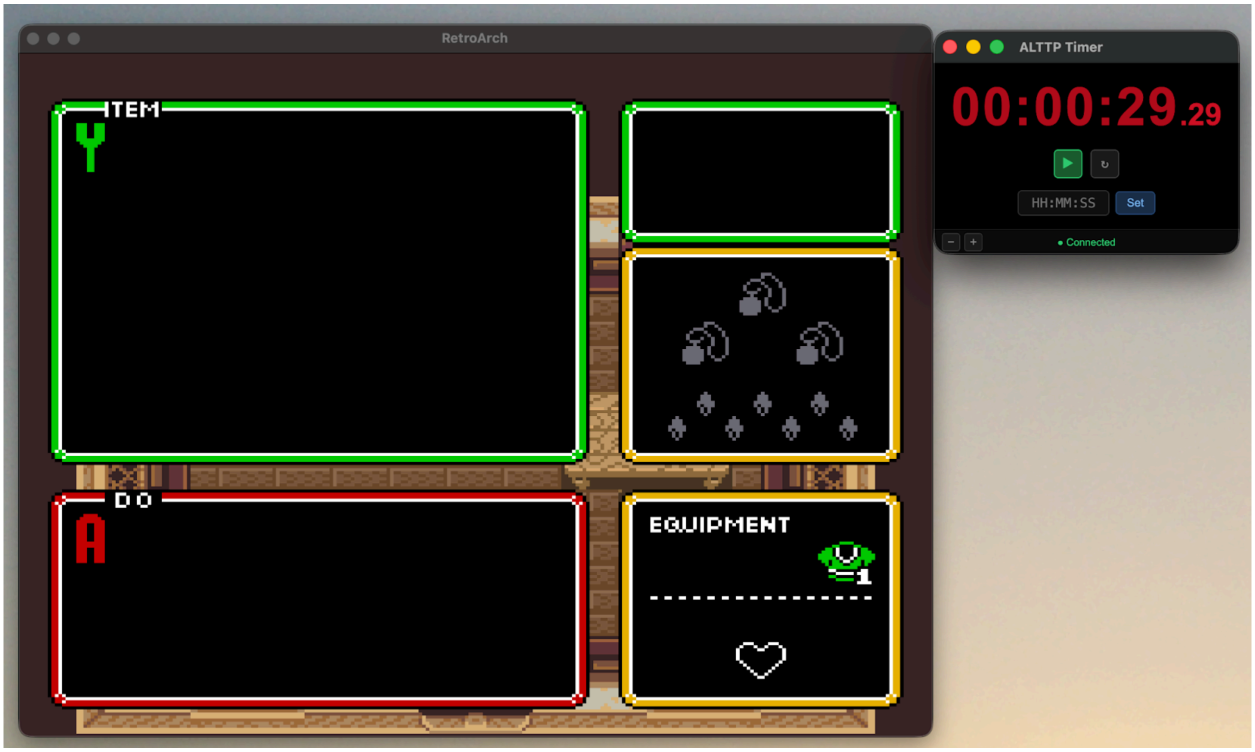
Auto-Start and Auto-Stop

Once connected, the timer automatically starts when the game transitions to active play (gamemode 0x07 or 0x09) and stops automatically when the Triforce room is entered (gamemode 0x19), which marks game completion.

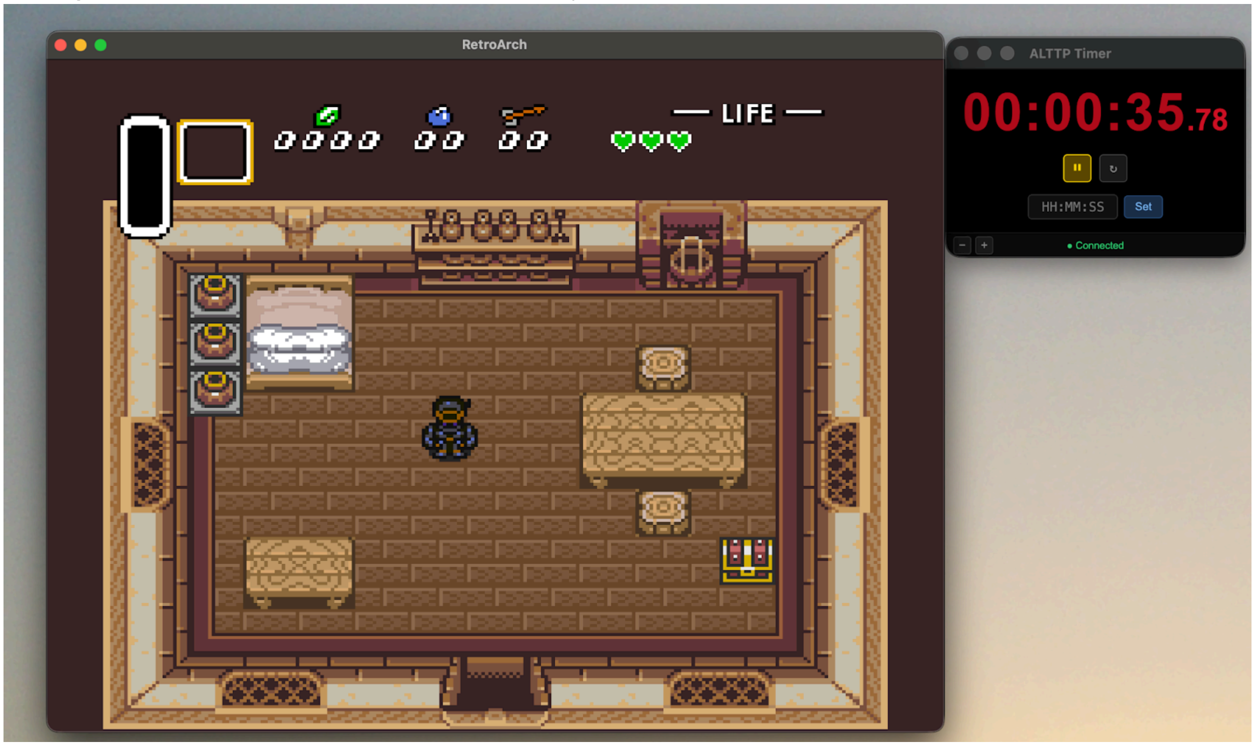
Pause

The timer can be paused when the game enters a menu state. It resumes when gameplay continues. You can also manually pause and resume using the pause button in the timer window.



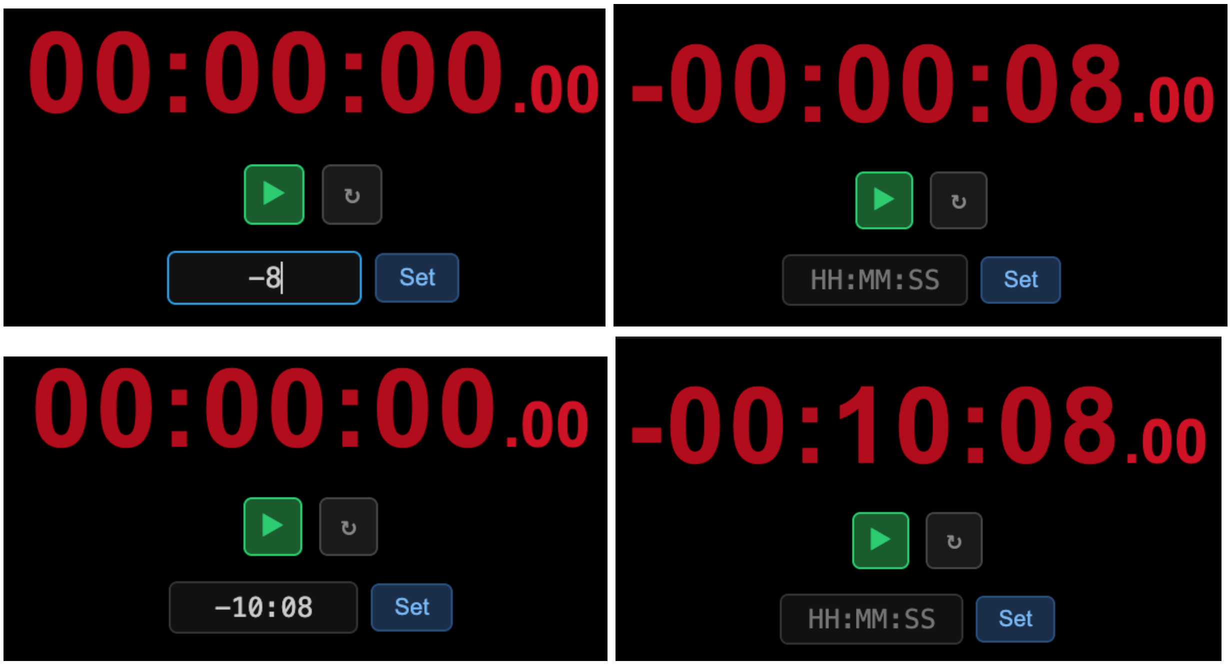

RetroArch
ITEM
DO
EQUIPMENT
ALTTP Timer
00:00:29.29
HH:MM:SS
Set
Connected

Exiting that menu, the timer will automatically start.



Negative Start Time

The timer supports a negative start time, which is useful for counting up from a defined offset. To set a negative start time, enter a negative value in the time input field (e.g., -8 for negative 8 seconds, or -10:08 for negative 10 minutes and 8 seconds) and press Set. The timer will start from that negative value and count up to zero and beyond.



Timer Color and Background

The number color and background of the timer are configured from the launcher. The color options are: blue, green, red, orange, purple, yellow, and pink. Background options match the item tracker: black, grey, white, and transparent.

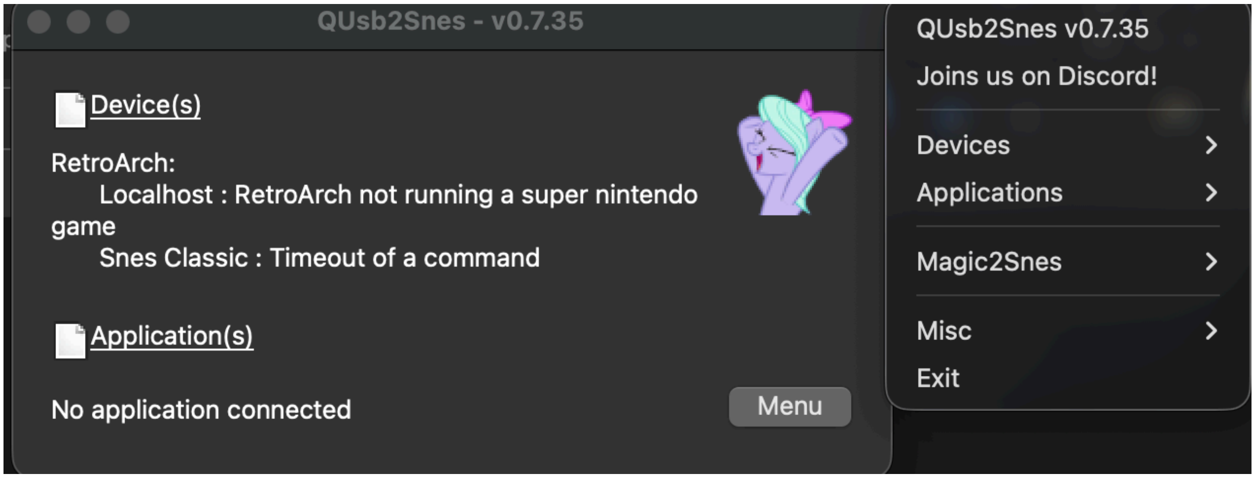
Required Software Installation

Install RetroArch v1.22.2 or higher. Then install exactly ONE WebSocket bridge — SNI or QUSB2SNES (both cannot run on the same port simultaneously).

QUSB2SNES Setup

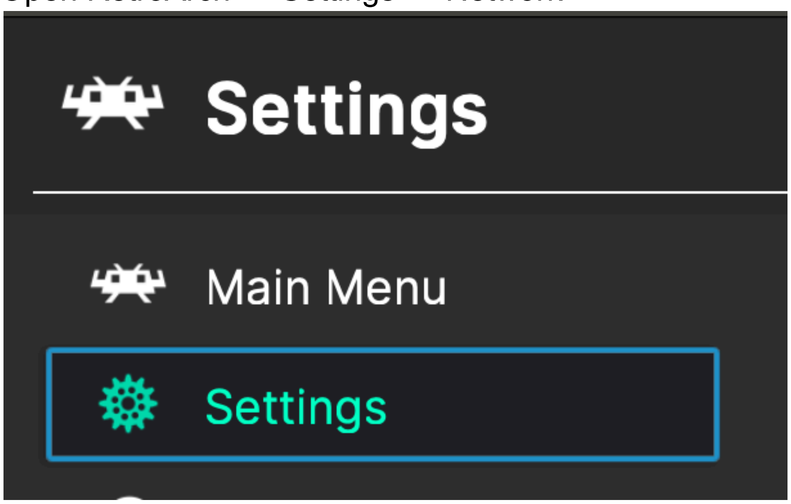
Install QUSB2SNES from: https://skarsnik.github.io/QUsb2snes/

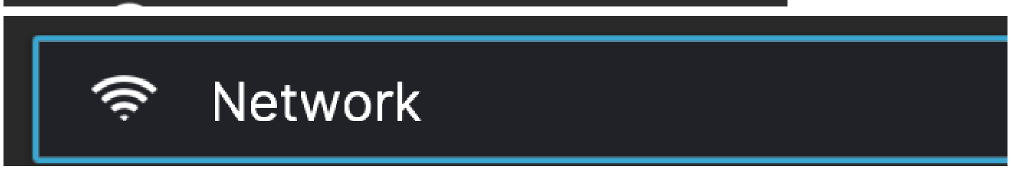
NOTE: Windows may require the Microsoft Visual C++ redistributable to run QUSB2SNES.



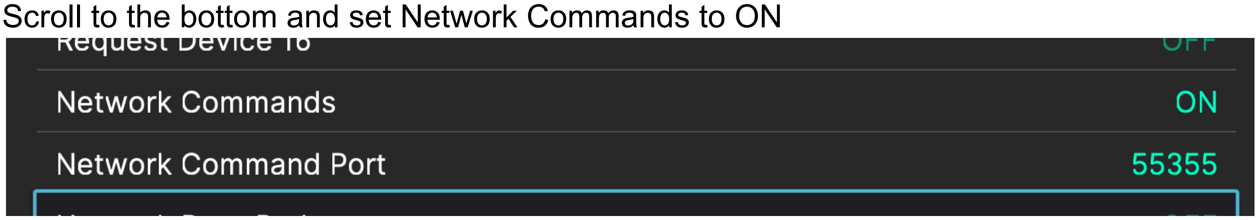
After launching QUSB2SNES, configure RetroArch to use the Snes9x core:

1. Open RetroArch → Settings → Network

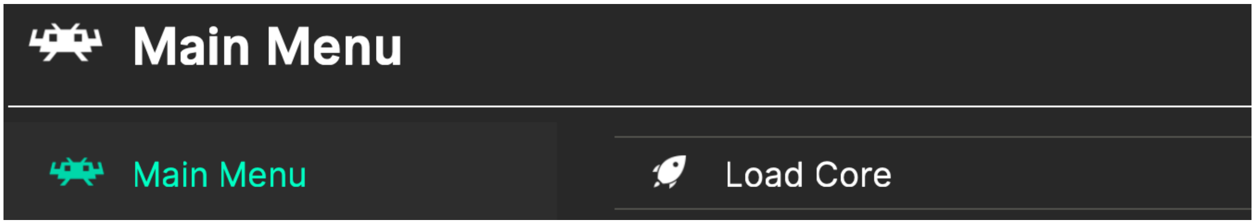




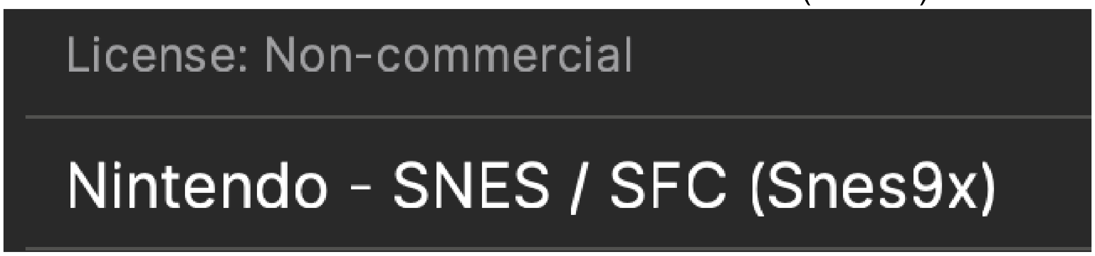
2. Scroll to the bottom and set Network Commands to ON



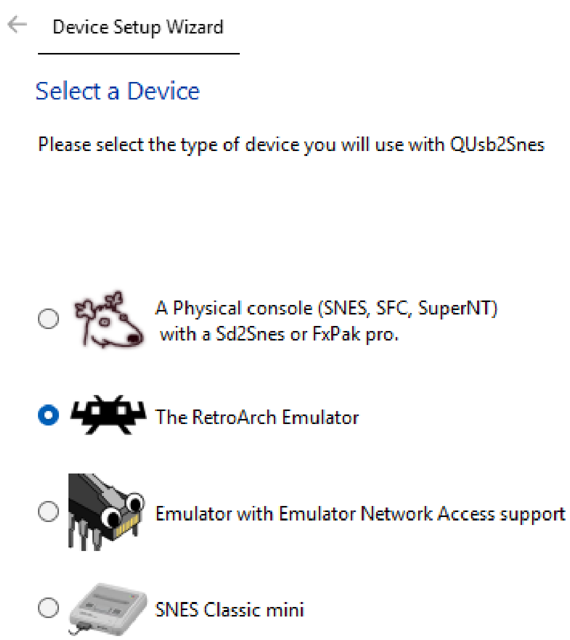
3. Return to Main Menu → Load Core → Download a Core



4. Search for SNES and install the Nintendo - SNES / SFC (Snes9x) core



NOTE
Windows will look like this when QUSB2SNES is launched
Select RetroArch

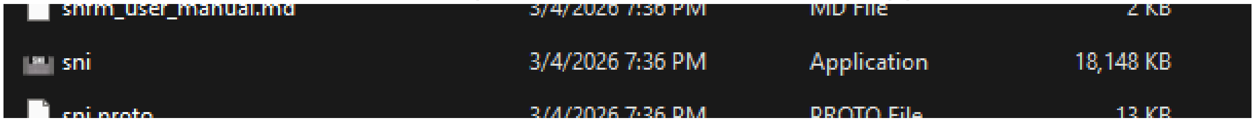


SNI Setup

Download SNI from: https://github.com/alttpo/sni/releases

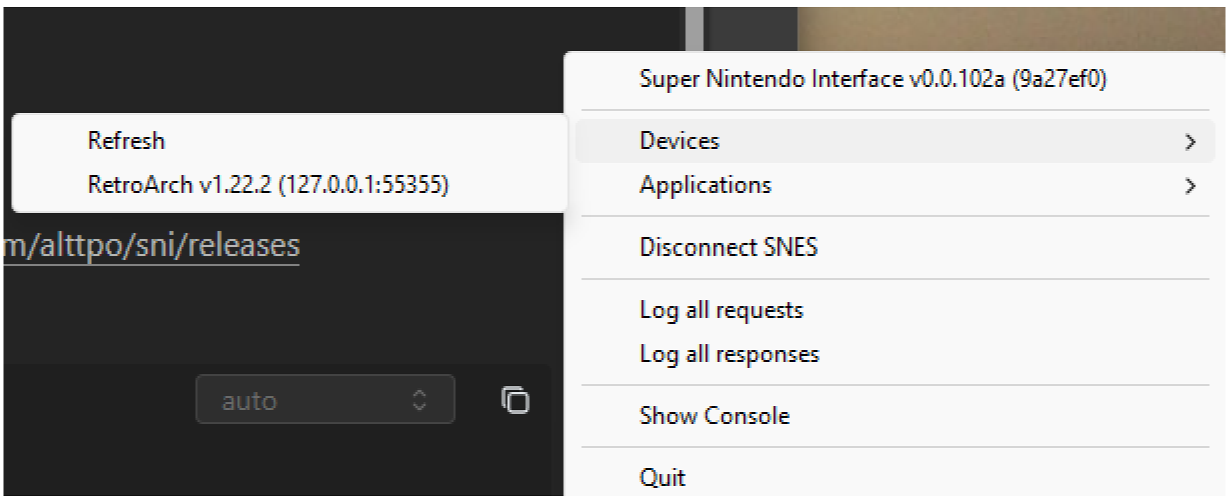
Windows

Download the Windows-amd64.zip, extract it, and run the sni application.



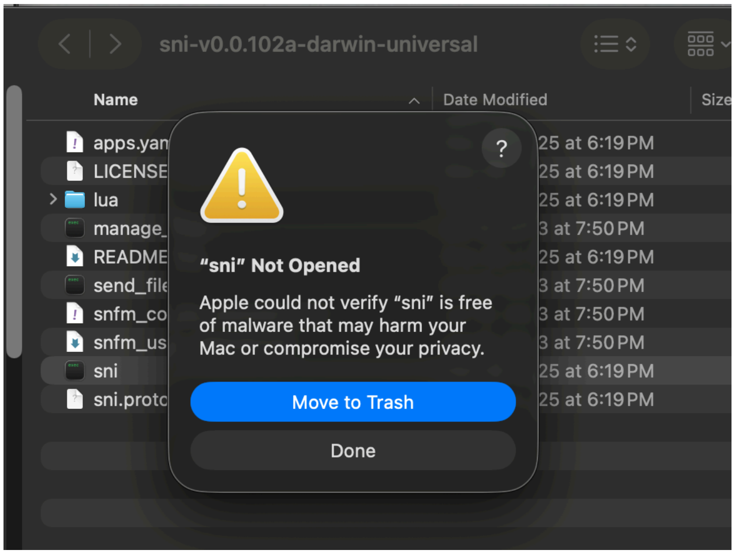
An SNI icon will appear in the system taskbar.

When RetroArch is running with a game loaded, the device will appear under Devices in the SNI menu.

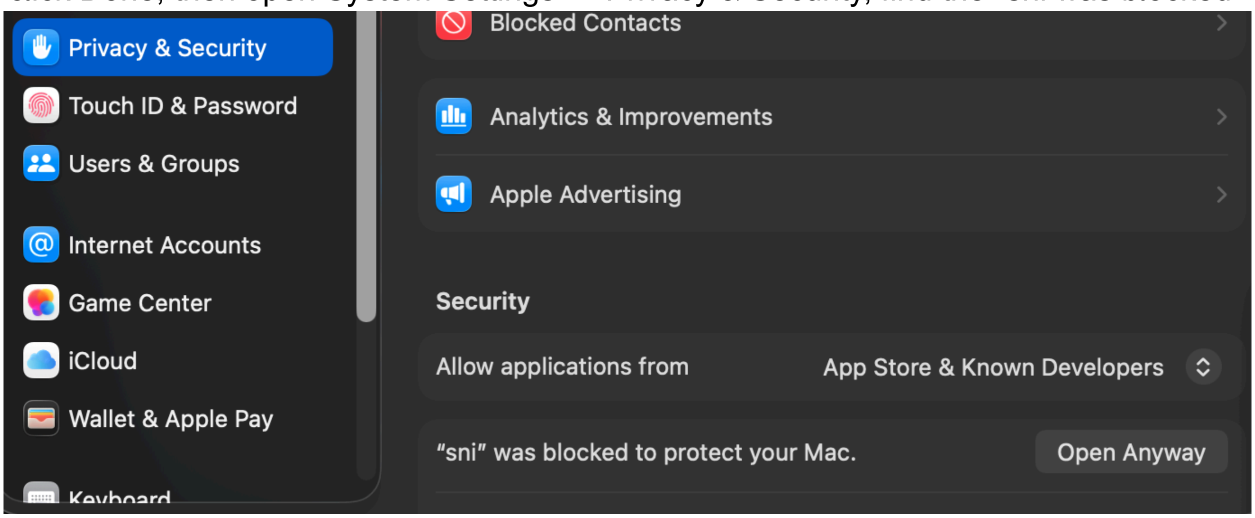


macOS

Download the darwin-universal archive, extract it, and double-click sni. macOS may show a security prompt



click Done, then open System Settings → Privacy & Security, find the "sni was blocked"



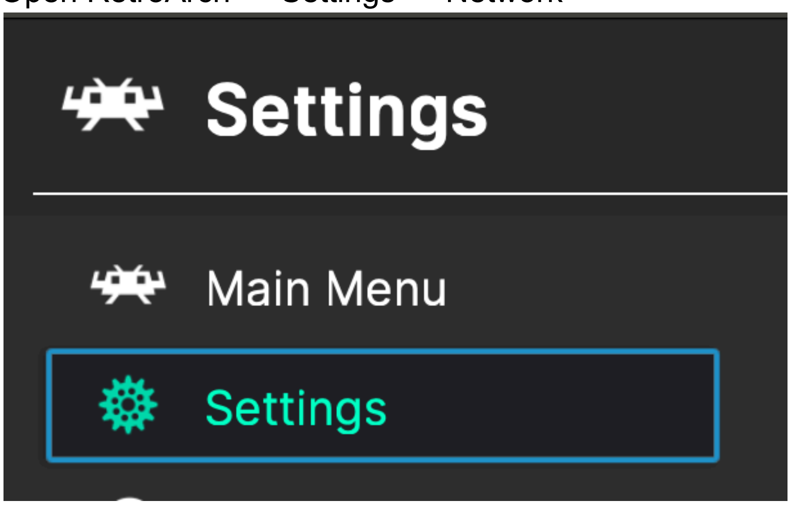
message, and click "Open Anyway."
A terminal window will appear — this is normal.

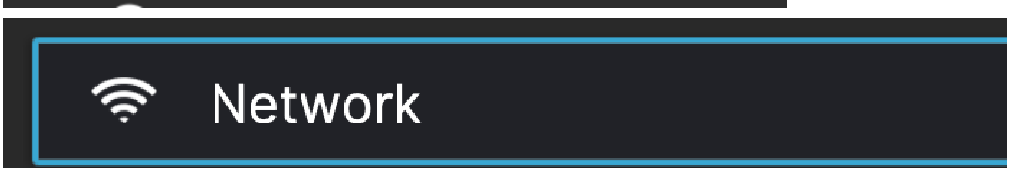
```
00000030  01 00 04 00 b7 ff b7 ff  2c 82 ab 98 00 80 af 98  |........,.......|
00000040  ff ff ff ff b7 ff 2c 82  2c 82 2c 82 00 80 d8 82  |......,.,.,.....|
2026/03/05 03:05:21.325433 detect: read {address:(FxPakPro $40ffb0 ExHiROM),size
:$50}
2026/03/05 03:05:21.341892 detect: read {address:(FxPakPro $40ffb0 ExHiROM),devi
ceAddress:(SnesABus $40ffb0 ExHiROM),size:$50} complete: score=21
00000000  01 8d 24 01 e2 30 6b 5c  9c b1 a1 ff ff ff ff ff  |..$..0k\........|
00000010  56 54 20 44 76 78 6e 44  36 62 4e 47 61 20 20 20  |VT DvxnD6bNGa   |
00000020  20 20 20 20 20 30 02 0c  05 00 01 00 8b 9f 74 60  |     0........t`|
00000030  01 00 04 00 b7 ff b7 ff  2c 82 ab 98 00 80 af 98  |........,.......|
00000040  ff ff ff ff b7 ff 2c 82  2c 82 2c 82 00 80 d8 82  |......,.,.,.....|
2026/03/05 03:05:21.342023 detect: read {address:(FxPakPro $40ffb0 SA1),size:$50
}
2026/03/05 03:05:21.358316 detect: read {address:(FxPakPro $40ffb0 SA1),deviceAd
dress:(SnesABus $01ffb0 SA1),size:$50} complete: score=4
00000000  a4 8f aa c1 7e 6b ad e0  02 d0 06 af 56 f3 7e d0  |....~k......V.~.|
00000010  17 af 57 f3 7e f0 03 9c  e0 02 a9 0c 85 4b a9 2a  |..W.~........K.*|
00000020  a6 1b f0 02 a9 14 85 11  6b a9 0c 85 4b a9 2a a6  |........k...K.*.|
00000030  1b f0 02 a9 14 85 11 22  eb 89 a4 38 e9 08 22 b9  |......."...8..".|
00000040  b2 a0 c9 a8 90 06 a9 00  8f 6d f3 7e 6b ff ff ff  |.........m.~k...|
2026/03/05 03:05:21.358419 detect: map mode = 30; masking off slow vs fastrom =
20
2026/03/05 03:05:21.358446 detect: detected mapping mode = LoROM
```

NOTE: SNI may work more reliably on macOS if launched after RetroArch and the game are already running.

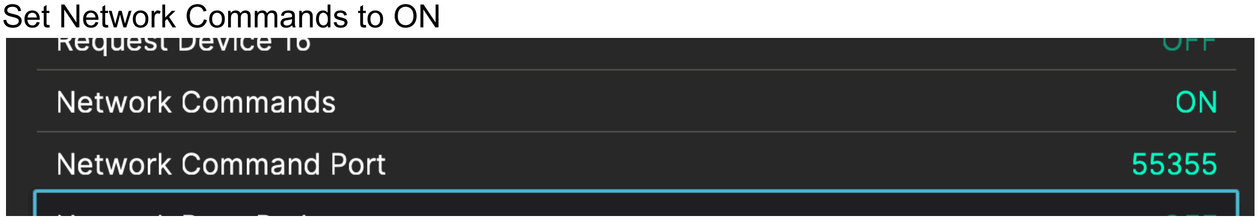
RetroArch Setup for SNI

5. Open RetroArch → Settings → Network

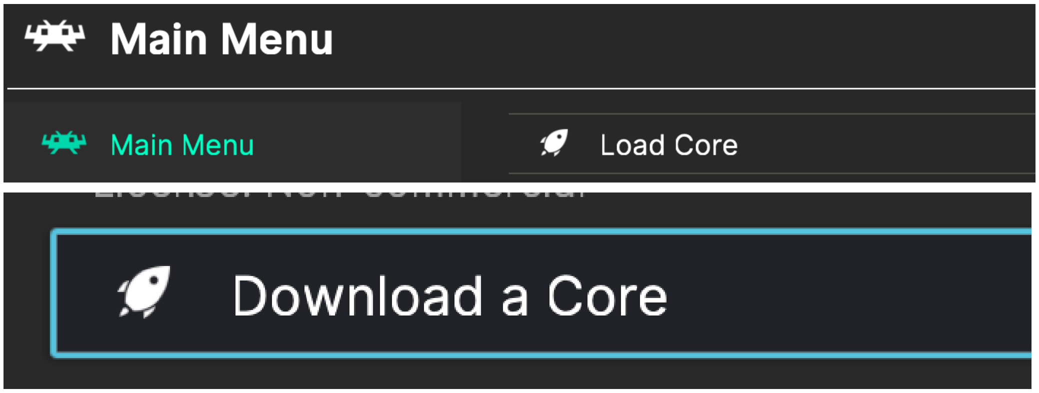




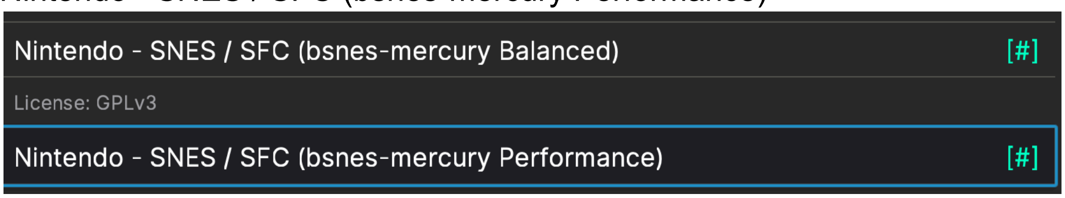
6. Set Network Commands to ON



7. Return to Main Menu → Load Core → Download a Core



8. Search for SNES and install either:
    a. Nintendo - SNES / SFC (bsnes-mercury Balanced)
    b. Nintendo - SNES / SFC (bsnes-mercury Performance)



NOTE: On macOS, test both Balanced and Performance cores to see which provides more stable autotracking.

Troubleshooting

Recommended Launch Order

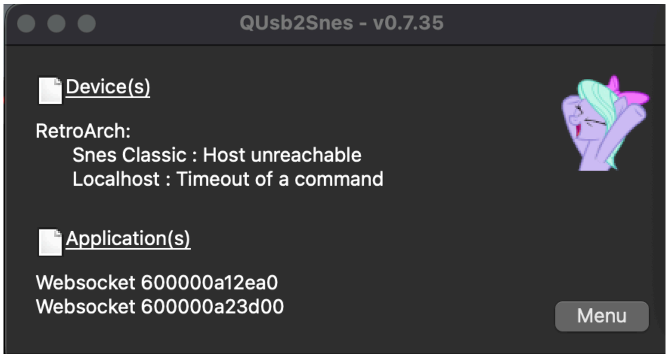
Almost all connection issues are resolved by following the correct launch order:

- Launch QUSB2SNES or SNI
- Open RetroArch
- Load the game and reach the title screen
- Launch the tracker (Items, Map, Timer)
- All windows should show "Connected" — then start your game

Tracker is Not Connecting

Check the WebSocket Bridge

Open the QUSB2SNES or SNI interface from the system taskbar.



If the tracker windows appear under Applications but no device is listed, RetroArch is not running or not detected. Make sure RetroArch is open with a game loaded.

WebSocket Bridge Not Running

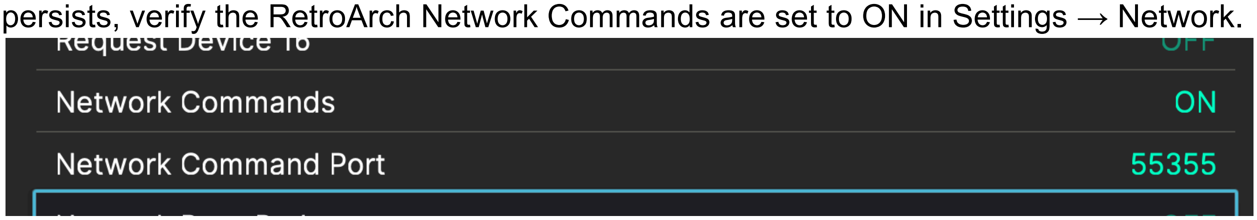
If neither QUSB2SNES nor SNI is running, the tracker cannot connect at all. Launch the appropriate bridge software first.

Still Not Connecting — Full Reset

- Close all tracker windows
- Close RetroArch
- Quit QUSB2SNES or SNI
- Relaunch in order: bridge → RetroArch (load game to title screen) → tracker
- Use the correct core for your bridge:
    - QUSB2SNES → Snes9x core
    - SNI → bsnes-mercury Balanced or Performance core

Nothing Works

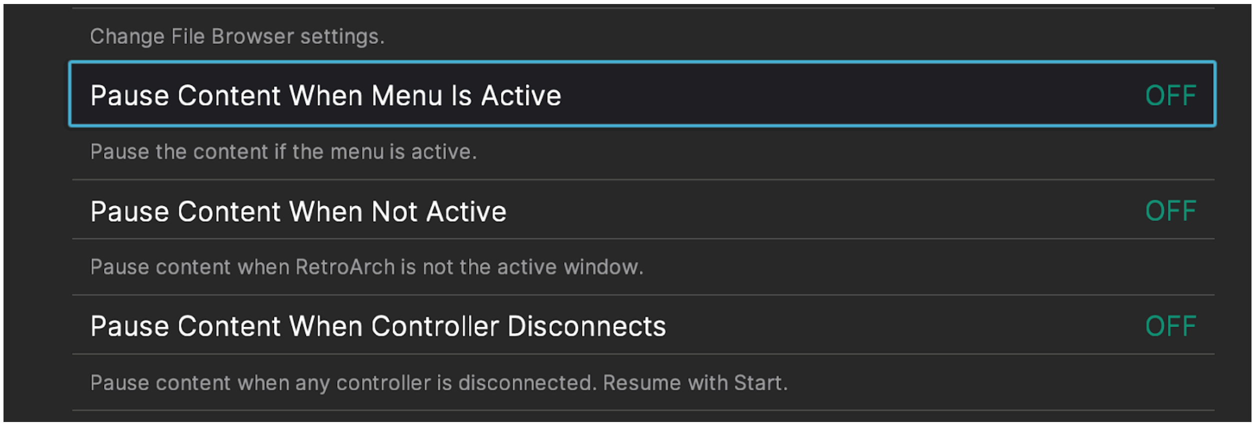
Uninstall and reinstall your WebSocket bridge (QUSB2SNES or SNI). If the issue persists, verify the RetroArch Network Commands are set to ON in Settings → Network.



Timer Not Working Correctly

If the timer starts and stops unexpectedly, RetroArch's pause-on-menu behavior may be interfering. To fix this:

- Open RetroArch → Settings → User Interface
- Set "Pause Content When Menu Is Active" to OFF
- Set "Pause Content When Not Active" to OFF
- Set "Pause Content When Controller Disconnects" to OFF



Reconnect Button

If a tracker window loses connection mid-session, click the reconnect button (↻) in the bottom-right corner of the Item Tracker or Timer window to re-establish the WebSocket connection without closing the window.

Known Limitations

- Floor item pickups in Desert Palace and Tower of Hera (the vanilla small key drop locations) are tracked via room flags that are also used by the small key counter. The tracker uses a cap on small key subtraction to keep item counts accurate, but these two floor items may briefly not register until a subsequent room state update.
- Sprite/drop data (used by some floor items in door shuffle modes) is not currently read by the tracker. Key Sanity mode is tested against standard Keysanity, Maps/Compasses, and Maps/Compasses/Small Keys shuffle settings.
- The publisher certificate for the Windows installer is not currently purchased, which causes the Windows SmartScreen warning on install. This does not affect the software's functionality or safety.

Acknowledgements

Special Thanks to Jedi Master T8ter for showcasing this tracker on stream and testing it. A thank you to LordHoell for indirectly providing ideas for the ItemTracker and for providing the idea on how to vibe code the map.
Finally a very special thank you to the ALTTPR community for the hard work with the randomizers and trackers.